

Gender Differences in Febrile Seizure–induced Proliferation and Survival in the Rat Dentate Gyrus

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Summary: *Purpose:* Febrile seizures are fever-associated early-life seizures that are thought play a role in the development of epilepsy. Seizure-induced proliferation of dentate granule cells has been demonstrated in several adult animal models and is thought to be an integral part of epileptogenesis. The aim of the present study was to investigate proliferation and survival of dentate gyrus (DG) cells born after early-life hyperthermia (HT)-induced seizures in male and female rats.

Methods: At postnatal day (PN) 10, male and female rats were exposed to heated air to induce seizures. Littermates were used as normothermia controls. Convulsive behavior was observed by two researchers. From PN11 to PN16, rats were injected with bromodeoxyuridine (BrdU) to label dividing cells. The number of BrdU-immunoreactive cells in the DG was counted at PN17 and PN66.

Results: At PN17, male as well as female HT rats had the same amount of BrdU-positive cells compared with controls. At PN66, significantly more BrdU-positive cells were left in HT females (53%) than in controls (44%, percentage of BrdU-positive cells at PN17), whereas no difference was found between HT males and male controls. The net result of proliferation and survival at PN66 was that female HT rats had the same number of BrdU-immunoreactive cells as controls, whereas male HT rats had 25% more BrdU-immunoreactive cells than did controls ($p < 0.05$).

Conclusions: Early-life seizures cause a sexually dimorphic cytogenic response that results in an increased population of newborn DG cells in young adult males, while leaving that of young adult females unaltered. **Key Words:** Febrile seizures—Hyperthermia—Hippocampus—Neurogenesis—Gender differences.

Febrile seizures (FSs) are the most common seizure type in young children. They occur in 3–4% of children between the age of 3 months and 5 years and are associated with a febrile illness. A number of genetic defects have been associated with a higher incidence of FSs, the so-called channelopathies, mutations in genes coding for ion channels, representing the largest group (1–3). Environmental factors have been found to increase the risk of developing FSs. For instance, in a population-based study, Lieberman et al. (4) found almost a fourfold increase in the risk of FSs, when maternal fever of $>100.4^{\circ}\text{F}$ (or 38°C) was measured during labor.

Although a number of factors contributing to FSs have been identified, still much debate exists on the consequences. Some studies suggest that FSs are benign, causing no damage to the child (5–7). However, this idea is

based on studies that are limited by the use of noninvasive techniques, the follow-up time, and often did not classify the types of epilepsy. In a prospective study, Tarkka et al. (8) failed to show structural or behavioral differences in children that experienced FSs when comparing them with age-matched, seizure-free controls. In contrast, Cendes et al. (9) found a 10-fold increased incidence of FSs (40%) among patients with mesial temporal sclerosis–associated temporal lobe epilepsy (TLE). Similarly, Camfield et al. (10) found an 8.8-fold increased risk for developing intractable epilepsy after prolonged FSs. The suggestion that FSs are causally related to the development of TLE is in line with the longstanding notion that “seizures beget seizures.” A number of animal models for FSs have been developed to study whether FSs induce epileptogenesis. Most animal models for FSs aim to mimic fever by increasing the body temperature, which can be achieved by exposing the animals to a heat source such as a microwave (11,12), infrared irradiation (13), or warm water (14). In the present study, we used the immature rat model developed by Baram et al. (15). The main advantage of this

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model is that it uses rats during a brain-development age comparable to that of a human child when it is most susceptible to seizures. Furthermore, the mortality and morbidity is low, which makes this model highly suitable for long-term follow-up studies.

None of the animal studies that used the model of Baram et al. (15) could conclude that hyperthermia (HT)-induced seizures cause TLE. They were unable to find histologic evidence for the presence of TLE. For instance, hippocampal cell death, a histopathologic hallmark of TLE, was not seen in these animals. Still, acute cell damage, evident by argyrophilic neurons, was present, but no change was seen in the total number of neurons (16). Chen et al. (17) showed that HT-induced seizures render the hippocampal network hyperexcitable by raising the depolarization current of neurons, thereby lowering the seizure threshold. Previously, daytime behavioral and electrophysiologic monitoring failed to demonstrate spontaneous seizures in adult rats after early-life HT-induced seizures. However, these animals had a lower threshold for seizures induced by administration of a subconvulsive dose of kainic acid (18). Recently, Baram et al. (19) showed that HT-induced seizures are able to elicit spontaneous nocturnal seizures. Another important finding is increased mossy fiber sprouting after HT-induced seizures, which might also contribute to increased excitability of hippocampal neurons (20).

In the past few years, altered neurogenesis has been proposed as a mechanism by which seizures modulate the hippocampal network. Several studies show that pilocarpine (21–23) or kainate-induced seizures (24,25), perforant pathway stimulation (25), or amygdala kindling (26,27) causes an increased proliferation of neurons in adult rats. Most of the newborn cells differentiate into neurons in the dentate granule cell layer and form connections with the CA3 area of the hippocampus and other dentate granule cells (21).

Gender differences in cell proliferation, differentiation, and survival (28,29) have been attributed to the role of hormones in neurogenesis (30). Evidence also exists of gender differences in seizure susceptibility. Males are found to be more susceptible to temporal lobe–like seizures because of high levels of testosterone (31). After administration of the convulsant *N*-methyl-D-aspartate, male rats show more severe seizure activity than female rats (32). Hence, the aim of the present study was to investigate proliferation of DG cells after early-life HT-induced seizures in male and female rats.

MATERIALS AND METHODS

Animals

Sprague–Dawley rats (Harlan, The Netherlands) were housed under standard conditions ($21 \pm 2^\circ\text{C}$ ambient temperature, a 12-h light/dark schedule, background noise provided by a radio, and food and water ad libitum). In

total, 21 male and 24 female rats were divided into the following four groups: (a) short-term normothermia (ST-NT), killed at postnatal day (PN) 17 ($n = 5$ males, $n = 5$ females); (b) short-term hyperthermia (ST-HT; $n = 6$ males, $n = 6$ females); (c) long-term normothermia (LT-NT), killed at PN66 ($n = 4$ males, $n = 6$ females); and (d) long-term hyperthermia (LT-HT; $n = 6$ males, $n = 7$ females). Rats were weighed daily from PN1 to 17, and once at PN66. All experiments were approved by the Animal Experiments Committee (DEC) of the University of Maastricht, The Netherlands.

Hyperthermia paradigm

HT was induced as described previously by Baram et al. (15), with minor modifications. In brief, PN10 rat pups were injected subcutaneously with 0.2 ml 0.9% saline to prevent dehydration and placed in a Perspex cylinder with a diameter of 10 cm (one rat/cylinder). An adjustable stream of heated air (50 – 52°C) was blown into the cylinder by a commercial hair dryer, placed at 50 cm above the rats, to raise the body temperature of the rat pups from a basal value of $\sim 35^\circ\text{C}$ to 41 – 42.5°C . Core temperatures were measured before and every 2.5 min during the HT treatment, with a rectal probe (K-type thermocouple bead probe connected to a ST-9612 digital thermometer; Velleman Components, Gavere, Belgium). When the core temperature of the rats reached 39.5°C , usually after 5–10 min, the temperature and volume of the air were adjusted so that a core temperature of 41 – 42.5°C was maintained for 30 min. If the core temperature exceeded 42.5°C , the rat was removed from the cylinder until the temperature dropped to 40 – 42.5°C . The occurrence of seizures was monitored behaviorally by two observers. The behavioral seizures were stereotyped and previously shown to correlate with EEG discharges in the hippocampus (15,18). Behaviorally, the seizures consisted of arrest of heat-induced hyperkinesis, followed by body flexion, and occasionally followed by clonic contractions of the limbs. The moment the rats showed body flexion was taken as the start of a seizure, whereas the end of the seizure was marked either by the end of the 30-min hyperthermia treatment or by regaining normal behavior. These time values were used to estimate the average seizure duration. Immediately after the 30-min treatment, the rats were placed in water (room temperature, RT) with their heads above the surface to regain their normal body temperature, after which they were returned to their mother.

Normothermia (NT) controls from the same litter as the HT rats were exposed to the same conditions, except that the stream of air was used to maintain the core temperature of the rats. At PN22, all pups were weaned and randomly housed two to three per cage. A schematic overview of the experimental design is presented in Fig. 1.

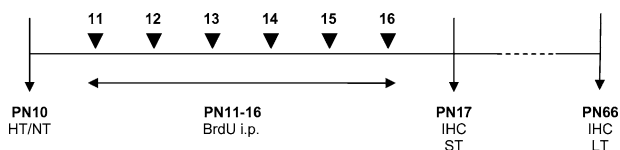


FIG. 1. Time schedule of the experimental design. At postnatal day (PN) 10, rats received a hyperthermia (HT) or normothermia (NT) treatment. From PN11 to 16, rats were injected twice daily, intraperitoneally (i.p.), with 25 mg bromodeoxyuridine (BrdU)/kg body weight. Immunohistochemical (IHC) staining for BrdU was carried out at PN17 (short-term, ST) or PN66 (long-term, LT).

BrdU immunohistochemistry

From PN11–16, all rats received twice daily (minimum 6 h apart) intraperitoneal injections of the thymidine analog 5'-bromo-2'-deoxyuridine [(BrdU; 25 mg/kg; Sigma, St. Louis, MO, U.S.A.), 2 mg/ml in 0.9% saline (pH 7.6)]. At PN17 or PN66, the rats received an overdose of pentobarbital (Nembutal), followed by perfusion fixation with 4% paraformaldehyde in 0.1 M phosphate buffer (PB, pH 7.6). After the brains were removed, they were postfixed in 4% paraformaldehyde/0.1 M PB for 48 h (4°C), and cryoprotected in 20% sucrose/0.1 M PB for 24 h (4°C). Coronal serial sections of 10 μ m were cut on a cryostat, and six sections per rat were mounted on Superfrost slides. The first and last sections were maximally 100 μ m apart from each other.

For bromodeoxyuridine (BrdU) detection, the sections were washed in TBS (0.1 M Tris-base, 0.15 M NaCl, pH 7.4), treated with TBS containing 0.6% H₂O₂ for 30 min, and washed again in TBS (2 $\times$ 10 min at RT, 1 $\times$ 10 min at 65°C). For DNA denaturation, the sections were incubated in 50% formamide/2 $\times$ SSC buffer (0.3 M NaCl, 0.03 M sodium citrate, pH 7.0) for 2 h at 65°C, rinsed in 2 $\times$ SSC buffer (RT), incubated in 2N HCl for 30 min at 37°C, and incubated in 0.1 M borate buffer (pH 8.5) for 10 min (RT). After this pretreatment, the sections were washed 6 times in TBS, incubated in TBS-TS (TBS containing 0.25% Triton X-100 and 3% normal donkey serum) for 60 min (RT), and then incubated overnight in primary anti-BrdU antibody (monoclonal mouse; Roche, Almere, The Netherlands; 1:800 in TBS-TS).

After rinsing the sections in TBS, they were incubated for 1 h in secondary biotinylated donkey anti-mouse antibody (Jackson ImmunoResearch Laboratories, West Grove, PA, U.S.A.; 1:400 in TBS-TS). The staining was visualized with a Vectastain ABC/Elite standard kit (Vector Laboratories, Burlingame, CA, U.S.A.), based on the avidin–biotin–peroxidase reaction, with diaminobenzidine as chromogen, and NiCl₂ as a signal enhancer. Finally, the sections were counterstained with 0.2% cresyl violet for 1 h, dehydrated, and coverslipped with DePeX. Sections of the NT and HT groups from both genders were processed simultaneously to minimize interassay variability.

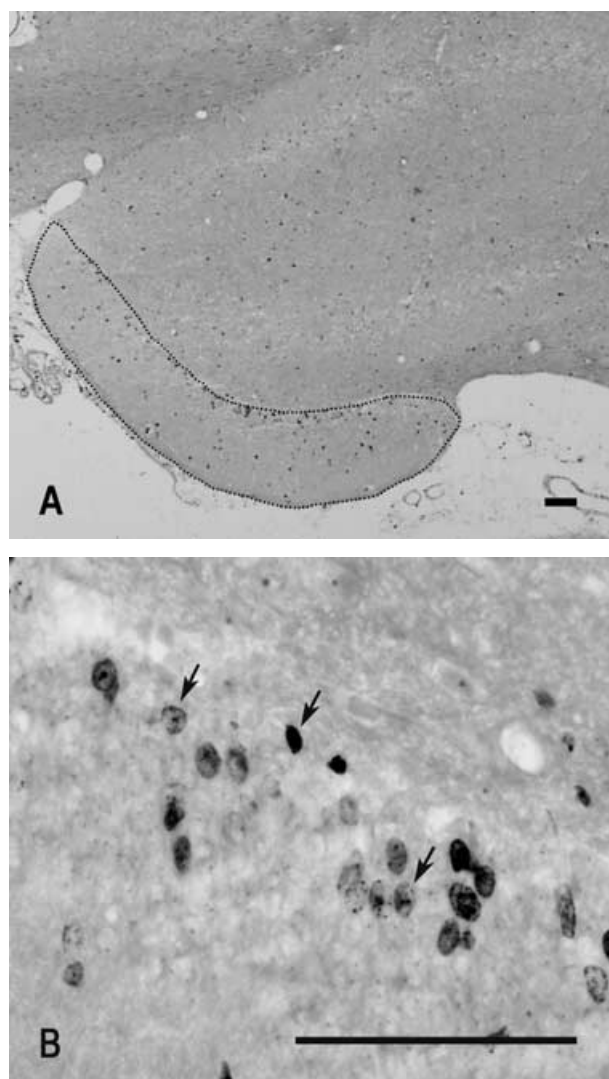


FIG. 2. Representative photomicrographs of a coronal section of the hippocampus of a 66-day-old male control rat. After immunohistochemical detection of bromodeoxyuridine (BrdU), the section was counterstained with cresyl violet. **A:** Dotted line, The delineation of the dentate gyrus (DG). **B:** Detailed picture of the granule cell layer of the DG. Cells that were uniformly or fragmentally stained (soccer-ball pattern) were counted (arrows). Scale bars, 100 μ m.

Quantitative analysis

Per rat, six coronal sections cut between bregma -2.12 and -2.30 mm were used as separate values for quantitative analysis of the BrdU staining. This analysis was carried out by using an Olympus BX51 microscope coupled to a computer supported by the Stereo Investigator program (MicroBrightField Inc., Williston, VT, U.S.A.). In each section, the left or right DG (randomly chosen) was delineated as depicted in Fig. 2A, the number of BrdU-positive cells in the delineated area was counted, and the surface of that area was calculated by the Stereo Investigator program. A cell was included if it was uniformly or fragmentally stained (see Fig. 2B) and had the morphologic

appearance of an oval or round nucleus. The reason for counting fragmentally stained cells is that the BrdU label dilutes with each cell division (33). The number of BrdU-positive cells per 100,000 μm^2 was calculated, and the cell counts of the HT groups were also expressed as percentage of NT (mean of 100%). Cell counts were performed at a $\times 500$ magnification by two observers blinded to the treatment status of the sections and showed no significant interobserver variation.

To avoid bias of potential changes due to the HT treatment or gender, the size of the DG (mean surface area of six sections) as well as the size of the cells (mean of 100 BrdU-labeled cells per rat) and cell density of the DG layer (DGL, total of BrdU-labeled and cresyl violet-stained cells) were measured by using the Stereo Investigator program (MicroBrightField).

Statistical analysis

Body weight measurements and growth curves were analyzed by using repeated measures analysis of variance (ANOVA) and Bonferroni's multiple comparison post hoc test. To analyze differences in body weight between treatment groups but within gender groups at each individual day, a Student's *t* test was used. Treatment, gender, and time effect on the number of BrdU-labeled cells, cell size, surface area of the DG, and cellular density were analyzed by using ANOVA with Bonferroni's post hoc test. Data are presented as mean $\pm$ SEM. Significance levels were set at $p < 0.05$.

RESULTS

In the present study, the effect of HT-induced seizures on proliferation and survival of DG cells was determined in immature male and female rats.

At the end of the HT phase, all HT rats displayed severe behavioral seizures, characterized by arrest of the heat-induced hyperkinesia, followed by body flexion and occasionally by clonic movements of the limbs. The average seizure duration in males (7.86 ± 0.81 min) was not significantly different from that in females (10 ± 1.54 min). None of the NT rats had behavioral seizures or showed abnormal behavior during their stay in the cylinder.

Body-weight measurements showed that the HT treatment had no overall effect on the growth curve (from PN12 onward, the slope of NT rats is not significantly different from that of HT rats) but did reduce weight gain in the first 1–2 days after HT (Fig. 3). ANOVA showed an interaction between Day and Treatment when body weights from PN10 to 16 were analyzed ($F_{6,246} = 5.42$, $p < 0.001$), but from PN12 to 16, this interaction was not present. Although pretreatment body weights of HT and NT rats were the same, the post-HT dip in weight gain resulted in a transiently reduced body weight of HT rats compared with their NT counterpart ($F_{1,41} = 13.32$, $p < 0.01$). From PN10 to 16, no overall effect of Gender was seen, and

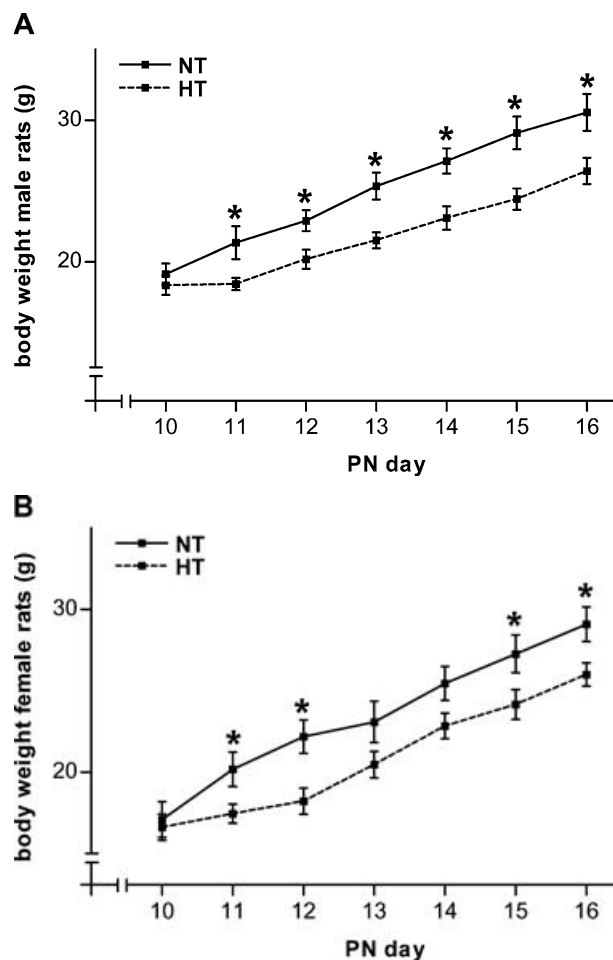


FIG. 3. Growth curves from postnatal (PN) days 10–16. Mean body weight of (A) male normothermia (NT, $n = 9$) and hyperthermia rats (HT, $n = 12$), and of (B) female NT ($n = 11$) and HT ($n = 13$) rats. Although pretreatment (PN10) body weights of NT and HT rats did not differ, posttreatment (PN11–16) body weights of HT rats were significantly reduced compared with their NT counterparts. From PN12 to 16, weight gain of NT rats was the same as that of HT rats (slopes of the NT and HT curves are not significantly different). Data are expressed as mean $\pm$ SEM. * $p < 0.05$ (difference between NT and HT, Student's *t* test).

no interaction between Gender and Treatment on body weight.

At PN66, the Treatment effect was no longer present, but then a Gender effect appeared ($F_{1,14} = 499.24$, $p < 0.001$), in which male rats weighed significantly more than female rats within each treatment group (post hoc; NT: male 279 ± 4 g vs. female 177 ± 4.43 g; $p < 0.001$; HT: male 282 ± 2.71 g vs. female 176 ± 4.42 g; $p < 0.001$).

To determine the effect of HT-induced seizures on proliferation and on survival of newborn cells, we labeled postseizure dividing cells with BrdU and quantified the number of BrdU-immunoreactive cells either shortly after labeling (i.e., at PN17) or 7 weeks after the last BrdU injection (i.e., at PN66). Figure 4 shows typical examples of the BrdU staining patterns found in the DG of the different

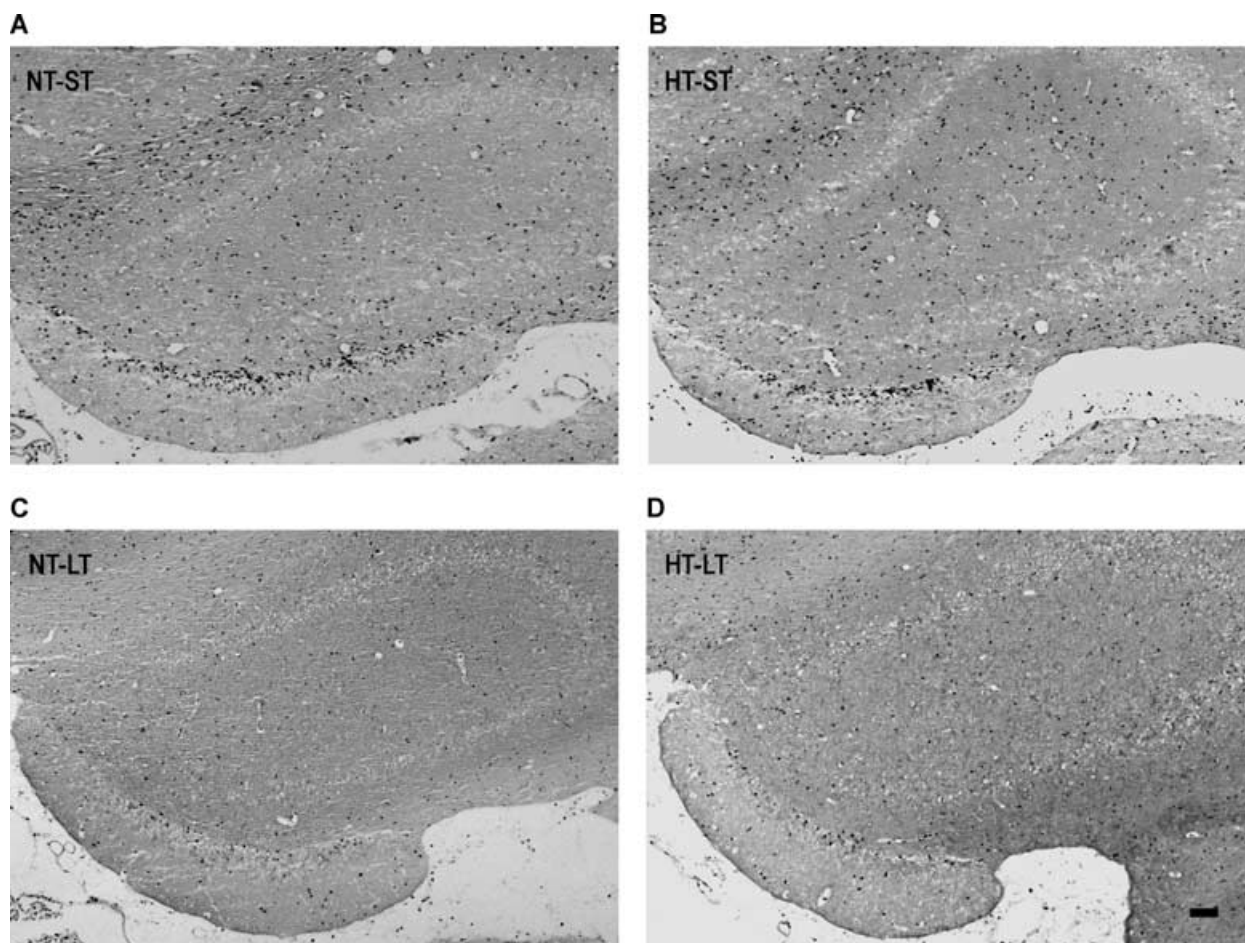


FIG. 4. Photomicrographs of bromodeoxyuridine (BrdU) immunolabeling in the dentate gyrus (DG) of the male rats. The sections were counterstained with cresyl violet to visualize the hippocampal structure. Fifty days after the last BrdU injection (LT), an unambiguous decrease in BrdU labeling is seen in the DG compared with labeling 1 day after the last injection (ST). BrdU staining patterns were similar in the female groups (pictures not shown). NT, normothermia; HT, hyperthermia; ST, short-term, PN17; LT, long-term, PN66. Scale bar, 250 μm .

treatment groups. They illustrate a clear decrease in BrdU labeling with increasing age (ST vs. LT). Only pictures of the male groups are presented, because male and female BrdU staining patterns looked alike.

BrdU quantification revealed that HT induces a gender-specific cytogenic response in the DG. To be able to compare BrdU counts between the present study and others, raw data of the number of BrdU-labeled cells/100,000 μm^2 are as follows: at PN17, male NT = 116.13 ± 3.22 ; male HT = 126.73 ± 4.6 ; female NT = 79.21 ± 2.35 ; and female HT = 71.46 ± 2.35 ; at PN66, male NT = 23.09 ± 1.79 ; male HT = 29.21 ± 1.66 ; female NT = 34.73 ± 2.15 ; and female HT = 37.78 ± 1.67 . At PN17, female NT and HT rats had significantly fewer BrdU-labeled cells per 100,000 μm^2 than their male counterparts (ANOVA, $p < 0.001$ for each treatment group). At PN66, no gender effect was noted. Notice the decrease in BrdU labeling at PN66 compared with PN17. ANOVA showed a significant time effect within each gender (ANOVA, $p < 0.001$ for

each gender). The data of the NT groups also are presented in Fig. 5.

To compare the HT effect between genders, the number of BrdU-labeled cells/100,000 μm^2 of the HT groups is presented as percentage of their control group in which the control groups have a mean of 100% (Fig. 6). At PN17, male and female HT rats had the same percentage of newborn cells as did their controls.

To compare gender differences in survival, we expressed the number of BrdU-positive cells at PN66 as a percentage of the number of BrdU-positive cells counted at PN17 (Fig. 7). From all the newborn cells at PN17 that were BrdU labeled during the first 6 days after treatment, 7 weeks later, 23% were left in HT males and 20% in NT males. In females, however, 53% survived in HT rats compared with 44% in NT rats (ANOVA with post hoc, $p < 0.05$).

BrdU quantification at PN66 (Fig. 8) shows the net result of proliferation and survival. About 25% more

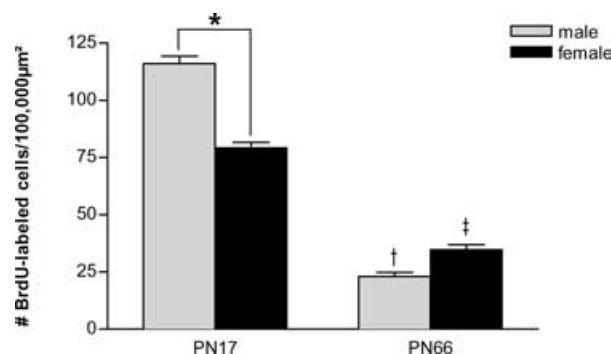


FIG. 5. Absolute number of bromodeoxyuridine (BrdU)-positive cells in the normothermia controls. At PN17, female rats had significantly fewer BrdU-positive cells than did male rats, whereas at PN66, this gender difference was no longer present. When comparing the number of BrdU-labeled cells at PN66 with that at PN17, a significant decrease is observed in the male and female group (PN17, male: $n = 5$ and female: $n = 5$; PN66, male: $n = 4$ and female: $n = 7$). Data are presented as mean $\pm$ SEM. *Male vs. female at PN17, $p < 0.001$. †Male PN66 vs. PN17, $p < 0.001$. ‡Female PN66 vs. PN17, $p < 0.001$.

BrdU-positive cells were found in the DG of male HT rats than in the DG of male NT rats (ANOVA with post hoc, $p < 0.05$). At this age, the number of BrdU-positive cells did not differ between female HT and female NT rats.

To avoid bias from potential changes in DG size, cell size and cell density of the DGL, associated with HT treatment or gender, these three parameters were compared between the groups. The results were as follows: at PN17, neither HT treatment nor gender had an effect on the surface area of the DG, cell size, or cell density; at PN66, HT treatment resulted in a significantly smaller DG surface area in female HT rats compared with female NT rats (ANOVA with post hoc, $p < 0.05$), whereas in males, HT treatment had no effect. Cell size and cell density were

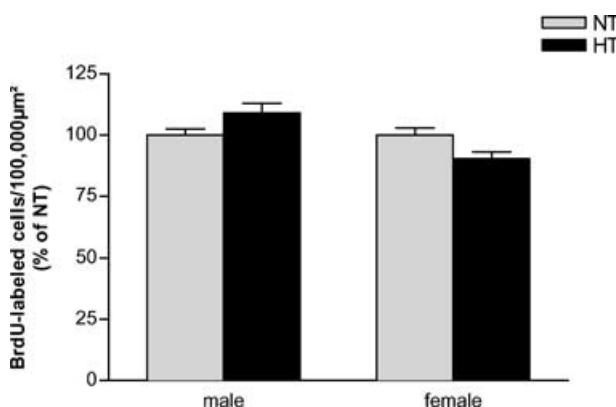


FIG. 6. Bromodeoxyuridine (BrdU) quantification in the dentate gyrus at PN17. After hyperthermia (HT) treatment, male and female rats showed the same percentage of BrdU-positive cells as their normothermia (NT) control rats (male: NT, $n = 5$ /HT, $n = 6$; female: NT, $n = 5$ /HT, $n = 6$). Data are presented as mean $\pm$ SEM.

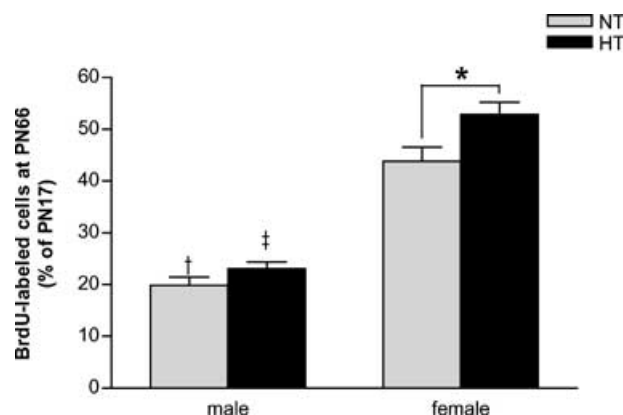


FIG. 7. Survival of newborn cells at PN66. Male hyperthermia (HT) rats showed no difference in surviving cells compared with male normothermia (NT) controls, whereas female HT rats had significantly more surviving cells than did female NT rats (male: NT, $n = 4$ /HT, $n = 6$; female: NT, $n = 6$ /HT, $n = 7$). Data are presented as mean $\pm$ SEM. †Male NT vs. female NT, $p < 0.001$. ‡Male HT vs. female HT, $p < 0.001$. *Female NT vs. female HT, $p < 0.05$.

not affected by treatment or gender at PN66. With respect to gender differences in growth, PN66 male rats had a larger DG than did females, irrespective of treatment. When comparing the values from PN66 with PN17, cell density did not differ between any of the groups, whereas the surface area of the DG was larger at PN66. This was irrespective of gender and treatment and was accompanied by an increased cell size at PN66.

DISCUSSION

One of the main features of epilepsy is a disturbed balance between excitation and inhibition, resulting in

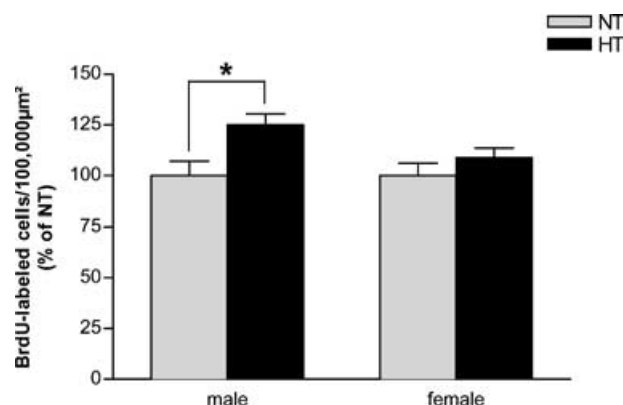


FIG. 8. Bromodeoxyuridine (BrdU) quantification in the dentate gyrus at PN66. The net result of proliferation and survival showed that male hyperthermia (HT) rats had 25% more BrdU-positive cells than did male normothermia (NT) rats, whereas the amount of BrdU-positive cells did not differ between female HT and NT rats (male: NT, $n = 4$ /HT, $n = 6$; female: NT, $n = 6$ /HT, $n = 7$). Data are presented as mean $\pm$ SEM. * $p < 0.05$.

hyperexcitability. A number of hippocampal abnormalities, such as mossy fiber sprouting, astrogliosis, and neuronal loss, may account for this hyperexcitability. In addition to these histopathologic findings, it was recently shown that seizures can alter the number of newborn DG cells (21–27). This observation has led to the hypothesis that seizure-induced neurogenesis may have a permanent effect on the constituents of the hippocampal network, thereby contributing to the process of epileptogenesis. Here we studied seizure-induced cytogenesis in the DG of developing male and female rats because, under normal physiologic conditions, neurogenesis is dependent on factors such as age and sex hormones (31,32).

The major findings from the present study are the following: (a) early-life seizures do not alter cytogenesis in the DG of male or female rats, (b) the survival of DG cells born after early-life seizures is, compared with that of littermate controls, increased in female rats and unchanged in male rats, and (c) the net result of proliferation and survival is that 8 weeks after treatment, HT males had 25% more BrdU-positive cells than did controls, whereas in females, no difference appeared between HT and control rats.

Gender differences in proliferation and survival of cells in normothermia controls

Sex hormones are known to play a role in proliferation and survival. Thus to study whether HT-induced seizures affect these processes, we first investigated gender differences in normothermia controls (Fig. 5). The analysis showed that at PN17, female rats had significantly fewer newborn cells than did male rats, whereas this difference was no longer present at PN66.

The lower proliferation in females found in the present study is in agreement with the results of Perfilieva et al. (29), who injected young naïve Sprague–Dawley rats with BrdU for 7 consecutive days. They also found that 1 day after the last injection, female rats had significantly fewer BrdU-labeled DG cells than did their male counterparts. Apparently a gender difference in basal proliferative rate of DG cells exists in immature rats. In line with the present study, they also evaluated the number of BrdU-labeled cells 30 days after the last BrdU injection and found no difference between males and females. Because PN17 females showed significantly fewer newborn cells than did males, and PN66 females had the same amount of BrdU-positive cells as PN66 males (Fig. 5), the survival (PN66 as a percentage of PN17; Fig. 7) was significantly higher in females (44%) than in males (20%). This result is similar to that found by Perfilieva et al. (29).

To our knowledge, gender differences in proliferation and survival have not been studied in rats of comparable age and at time intervals similar to those used in the present study. Although some information exists about the influence of gender on proliferation in adults, the number

of studies is limited. A few studies, such as that by Tanapat et al. (28), found a transient increase in the number of newborn cells in adult female rats. This was attributed to increased levels of estrogen because estrogen-treated ovariectomized rats showed an increased proliferation. Because the rats in the present study were treated during the first 2 weeks of their lives and therefore had not reached sexual maturity when first analyzed at PN17, it is unlikely that a higher estrogen level in the females was the cause of the gender difference in proliferation found here. Conversely, sex differences in estrogen-receptor density, or perhaps binding capacity, might explain our data. However, O'Keefe et al. (34) found no sex differences in the concentration of estrogen receptors or in estradiol-receptor binding in young Sprague–Dawley rats. An alternative explanation for the gender difference in proliferation comes from a study by Pang et al. (35). They found that serum concentrations of testosterone from PN1 to 10 were ~3 times higher in male rats than in females, whereas serum levels of estrogens did not differ between the genders. Because neonatal administration of testosterone has been found also to increase postnatal cell proliferation (36), this might explain the higher proliferation in males compared with females in our study.

Conversely, the time point at which survival was analyzed does coincide with the age at which the rats had attained sexual maturity (PN66). Although the molecular mechanisms of survival of newborn cells are not well understood, gender differences suggest that sex hormones are involved in this process as well. For example, estrogen has been found to exert a neuroprotective effect (37,38). Therefore the increased survival in females may occur as a result of an increase in estrogen levels and consequently a reduced cell death.

Absence of gender differences in seizure-induced proliferation

Because seizures can alter proliferation and because, as mentioned earlier, a gender difference is found in basal proliferative rate, we studied the effect of HT-induced seizures on proliferation in both male and female rats. We found no effect of HT-induced seizures on proliferation in either of the two genders (Fig. 6). It is known that postnatal neurogenesis can be modified by many factors. Here, we discuss several factors that may have particular importance for the present study.

First, factors related to the treatment, such as seizure duration, weight loss, and stress, may affect the BrdU cell counts. The observation that HT-induced seizures did not alter the birthrate of DG cells 1 week after the seizures (PN17) is in agreement with the results of a previous study by Bender et al. (20). The authors suggested that the duration of the HT-induced seizures is insufficient to provoke DG cell proliferation in the immature hippocampus. This idea was based on the observation that

long-lasting (>120 min) KA-induced seizures increased the number of newborn cells when compared with littermate controls, and HT-induced seizures were unable to change the number of proliferating cells. However, a difference in newborn cells was present in the control groups (i.e., the control group that was used to compare with the KA-treated animals had fewer BrdU-labeled cells than the control group that was used for comparison with the HT-treated animals). The HT- and KA-treated animals, in contrast, showed the same amount of BrdU-labeled cells. Weight loss also may have affected our results. For instance, food deprivation resulting in reduced weight gain is associated with decreased proliferation (39). In line with the results of Stern et al. (40), we found that HT treatment caused a transiently reduced body weight. This was probably due to a reduced food intake during the first 24 h after treatment, because from PN12 on, HT and control rats showed a similar weight gain, and on PN66, a difference in body weight was no longer seen between HT and control animals. More important, this effect was similar in male and female rats, and at this early age, no difference in body weight was found between male and female controls, which is in agreement with findings of Perfilieva et al. (29). We thus have no indications that HT-induced weight loss affected our results. Finally, stress is another well-known factor able to influence the proliferative rate. Stress is known to decrease proliferation in the adult rat (41,42). The stress in the present study might derive from the treatment itself, for instance, from maternal separation. Although we cannot exclude that isolation from the mother played a role in the neurogenesis process in these pups, for several reasons, it is unlikely that the stress induced by isolation affected our findings. First, the separation during the NT or HT treatment only lasted 30 min, whereas in most studies pups and mothers are separated for 3 to 24 h. In these studies, it seems that the lack of tactile stimulation by the mother and milk deprivation alter physiologic responses [for review, see (43)]. During the isolation, we frequently measured body temperature, which involved tactile stimulation, and reduced dehydration by injecting saline before separation. Moreover, Severino et al. (44) studied gender differences in stress response at different stages in the animal's lifespan after neonatal handling. They handled the animals from PN1 to 10 for 1 min/day. At PN11, they found no difference in the stress response between males and females. Additionally, and most important, male and female controls and HT-treated animals were exposed to the same level of stress due to isolation. In addition to maternal separation, seizures also may have caused some stress in this model. Liu et al. (45) showed that suppression of hippocampal neurogenesis in immature rats is associated with the number of seizure episodes and glucocorticosteroid levels. We induced HT seizures only once and found no change in proliferation. This is in agreement with the results of Liu et al., who also found no

change in the number of newborn cells after one seizure episode. To summarize, treatment effects such as seizure duration, weight loss, and seizure-induced stress do not seem to have affected proliferation in this model.

A second important factor that may have affected our results is the method applied. In the present study, proliferating cells were labeled with BrdU for 6 consecutive days after the NT or HT treatment. Thus evaluation of the number of BrdU-positive cells 1 day after the last injection (1 week after the treatment) cannot exclude that some newborn cells had already died. It is suggested by Hayes and Nowakowski (33) that a more accurate estimation of pure proliferation can be given when injecting a single dose of BrdU immediately after treatment, followed by evaluation of the number of BrdU-positive cells within 24 h. However, Bender et al. (20) followed this suggestion and did not find differences in proliferation after HT-induced seizures either. The absence of an altered proliferation rate after HT-induced seizures in the present study might derive from increased cell death accompanying increased proliferation. However, like Bender et al., others have used the HT model to study cell death after seizures and failed to detect increased cell death (16,20,46). We therefore conclude that indeed no change in proliferation occurred in our study. To our knowledge, it is not yet known whether gender differences in cell death exist in the HT model for early-life seizures. Furthermore, to avoid bias from potential differences in DG size, cell size, and cell density, we analyzed whether gender or treatment affected these parameters. We found no overall gender or treatment effect on the DG size, cell size, or cell density at PN17.

Gender differences in survival of cells born after HT-induced seizures

This is the first report that has quantified the survival of DG cells born after HT-induced seizures in immature male and female rats. At PN66, male controls had 20% and HT males had 23% BrdU-labeled cells compared with those at PN17. HT-induced seizures also increased the survival of newborn cells in females, because at PN66, female controls had 44%, whereas HT females had 53% BrdU-labeled cells compared with those at PN17 (Fig. 7). Thus HT-induced seizures hardly affected the survival of newborn DG cells in males and significantly increased that in female rats. At PN66, cell size and cell density were not affected by gender or treatment. The DG size was not affected by treatment in males; however, female HT rats had a 14% smaller DG than their NT counterparts. This result is somewhat difficult to explain but is probably not due to the treatment because such a difference was not found at PN17 or in males. Most important, cellular density and cell size were the same in both groups, and by visual inspection of cresyl violet-stained sections, we did not find any evidence of damage in the DG of the female HT rats. The most likely explanation therefore is

utilization of sections from different levels, with sections from NT females taken somewhat closer to bregma -2.30 than those of female HT rats. The main limitation of this finding might be that the sample size in the HT group is not as accurate as in the NT group.

Our results showed substantial gender differences in survival. As described earlier, survival of newborn cells in female controls was significantly higher than that in male controls. In addition to the gender-related difference in survival, HT-induced seizures caused an even higher increase in the survival rate in females, although they did not affect the survival rate in males. This might be explained by the fact that seizures are known to change the level of reproductive hormones [for review (47,48)], such as an increase in estrogen levels. Recently, Baram et al. (19) showed that adult rats (PN70) develop nocturnal spontaneous seizures after early-life HT-induced seizures. So these spontaneous seizures, if present in our rats, may be responsible for the sexually dimorphic survival rate. At PN17, we did not find a different effect of the HT treatment between the two genders. This suggests that the HT-induced seizures at PN10 are not able to affect the levels of sex hormones as can spontaneous seizures.

Consequences of altered proliferation and survival after HT-induced seizures

The net result of proliferation and survival was that 8 weeks after HT-induced seizures, female HT rats had similar amounts of BrdU-labeled cells compared with controls, whereas male HT rats had 25% more BrdU-labeled DG cells than did controls (Fig. 8). Porter and colleagues (49) recently assessed the survival of cells born after pilocarpine-induced status epilepticus in PN20 rats, using BrdU injections to label cells dividing 4, 6, and 8 days after seizure. Three weeks after the last BrdU injection, they also found significantly more BrdU-labeled DG cells in status epilepticus rats than in controls. Further characterization of these surviving cells revealed that $\sim 90\%$ coexpressed the neuronal marker NeuN and thus were neurons. They also found that the percentage of BrdU-labeled cells that coexpressed NeuN did not differ between controls and rats that had experienced seizures. So they concluded that status epilepticus did not alter the neuronal fate of the surviving cells. The observation that seizure-induced cytogenesis in immature rats mostly results in the birth of neurons confirms previous studies (23,50). A recent study by Dayer et al. (51) suggests that these surviving neurons are very stable and may permanently replace neurons that are integrated into circuits. Previous studies using the HT model in male rats failed to detect significant hippocampal neuronal dropout at 4 weeks after seizures (16,20,46), resulting in the same total number of neurons in controls and seizure rats. Our own data failed to demonstrate a treatment effect on cell density. Thus the increased number of BrdU-labeled cells we observed 8 weeks after HT-induced

seizures in male rats suggests that cell death did occur, but dead cells have been replaced by newborn cells.

Recently, a review article by Shapiro and Ribak (52) described a change in the migration pattern of newborn DG cells after seizures. The so-called basal dendrites, which normally retract early in DG cell development, fail to do this after seizures, resulting in an ectopic location of these cells in the hilus. This idea was based on the results of several studies, in which it was shown that seizures caused an increased number of newborn granule cells that resided in the hilus. These ectopic cells, although present in small numbers, are suggested to play a role in hippocampal hyperexcitability seen in epileptic rodents. The functional consequences of the increase in cells in the male HT rats seen in the present study remain to be elucidated but may be resolved in part by determining the identity and migration of these newborn cells.

To summarize, it was shown that HT-induced seizures do not change proliferation of DG cells in male or female rats. Conversely, survival of these newborn cells was higher in female than in male rats. The net result is an increased population of newborn DG cells in young adult males, while leaving that of young adult females unaltered, which might contribute to gender-related differences in seizure susceptibility.

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